DMPC for Multi-Drone Motion Planning

with Dynamic Goal Locations

Distributed Model Predictive Control + gym-pybullet-drones

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The Problem

Existing DMPC approaches assume:

- Static goal locations for all agents
- Pre-defined end positions known at start
- No adaptation to moving targets

Real-world scenarios need:

- Dynamic targets (moving objects, changing missions)
- Real-time trajectory adaptation
- Collision-free paths with moving goals

Our Solution

Extending DMPC to handle dynamic goals using:

1. **gym-pybullet-drones** simulation environment
    - Physics-based quadrotor dynamics
    - Real-time visualization
    - Validated drone models (Crazyflie 2.0)
2. **Moving goal adaptation**
    - Online replanning for goal updates
    - Event-triggered re-optimization
    - Maintain collision avoidance guarantees

Technical Approach

Core Components:

- **Bézier curve parameterization** for smooth trajectories
- **On-demand collision avoidance** (50% faster than BVC method)
- **Event-triggered replanning** for disturbances & goal changes
- **Distributed QP optimization** solving at 5 Hz planning rate

Key Innovation: Adapting the optimization problem to track moving targets while preserving collision-free guarantees

Expected Impact

Simulation Results (from base paper):

- ✓ 90%+ success rate with 30 agents in dense environments
- ✓ 50% reduction in travel time vs. traditional methods
- ✓ Real-time capable (20 Hz control with 20 drones)

Our Extension:

- Dynamic goal tracking in realistic physics simulator
- Demonstrate robustness to moving targets
- Validate scalability with gym-pybullet-drones

Next Steps

1. Implement base DMPC in gym-pybullet-drones
2. Extend cost function for moving goal tracking
3. Test with varying goal dynamics (static → moving)
4. Benchmark performance & collision rates

Goal: Enable autonomous drones to handle dynamic mission objectives in real-time

Questions?

Reference Paper:

Luis et al., "Online Trajectory Generation with Distributed Model Predictive Control for Multi-Robot Motion Planning"

IEEE Robotics and Automation Letters, 2020

https://ieeexplore.ieee.org/document/8950150